

Parsing BLAST XML with `Bio.Blast.NCBIXML`

Why BLAST XML parsing is needed?

BLAST results are usually returned in **XML format**, which is:

- Machine-readable
- Rich in details (scores, identities, E-values, alignments)

To extract useful info, we **parse XML**.

BLAST XML Structure (Concept)

BLAST XML contains:

- One or more **BLAST records**
- Each record has multiple **alignments (hits)**
- Each alignment has **HSPs** (High Scoring Pairs)

Biopython XML Parser

```
from Bio.Blast import NCBIXML
```

Example: Parsing BLAST XML

```
from Bio.Blast import NCBIXML

with open("blast_result.xml") as handle:
    blast_records = NCBIXML.parse(handle)

    for record in blast_records:
        for alignment in record.alignments:
            for hsp in alignment.hsps:
                print("Hit ID:", alignment.hit_id)
                print("Length:", alignment.length)
                print("E-value:", hsp.expect)
                print("Identity:", hsp.identities)
                print("Alignment:")
                print(hsp.query)
                print(hsp.match)
                print(hsp.sbjct)
                print("-" * 40)
```

Important Fields You Should Know

Field	Meaning
hit_id	Sequence ID in database
hit_def	Description
identities	Exact matches

Field	Meaning
align_length	Alignment length
expect	E-value
score	Alignment score

Interview tip: **HSP = best local alignment region**

Local BLAST Installation (Conceptual)

What is Local BLAST?

Running BLAST **on your own computer** instead of NCBI servers.

Why use Local BLAST?

Online BLAST	Local BLAST
Easy	Fast
Limited	No limits
Manual	Automated
Small data	Large datasets

Basic Steps (Awareness level)

1. Download **BLAST+** from NCBI
2. Install on system
3. Download databases (nt, nr, swissprot)
4. Run BLAST via command line

Example command:

```
blastn -query query.fasta -db nt -out result.txt
```

You are **not expected to memorize commands** as a beginner, but concept is important.

Filtering BLAST Results

Why filtering is important?

BLAST gives **many hits**, but not all are meaningful.

Common Filtering Criteria

Filter	Why
E-value	Remove random matches
Identity %	Keep high similarity
Query coverage	Ensure sufficient alignment
Alignment length	Avoid tiny matches

Example: Filter by E-value

```
if hsp.expect < 1e-5:
    print("Significant hit")
```

Example: Filter by Identity %

```
identity_percent = (hsp.identities / hsp.align_length) * 100

if identity_percent > 70:
    print("High similarity hit")
```

Industry standard: **E-value < 1e-5**

Protein BLAST & BLOSUM Matrices

Why Protein BLAST is different?

Protein sequences:

- Tolerate substitutions
- Evolution is slower
- Chemical similarity matters

What is BLOSUM?

BLOSUM (BLOcks SUBstitution Matrix) assigns scores based on:

- How frequently amino acids substitute in evolution

Example:

- Leucine ↔ Isoleucine → high score
- Glycine ↔ Tryptophan → low score

Common BLOSUM Matrices

Matrix	Use
BLOSUM62	General purpose (default)

Matrix	Use
BLOSUM80	Closely related proteins
BLOSUM45	Distant proteins

Protein BLAST with Biopython

```
from Bio.Blast import NCBIWWW

result_handle = NCBIWWW.qblast(
    program="blastp",
    database="nr",
    sequence="MVLSPADKTNVKAAWGKVGAHAGEYGA"
)
```

BLASTP automatically uses BLOSUM matrix (usually BLOSUM62)